

affect the economic tradeoffs in retirement. It is the beginning of a discussion rather than the end of a process.

Thus, for example, one glaring omission from all these simple stories is the precise role of life insurance in maximizing the estate value. After all, if the retiree is sufficiently well off so that he is never likely to exhaust his nest egg regardless of his spending rate, then life insurance is likely to be part of the optimal portfolio, at the very least for estate planning purposes. Likewise, the exact role of personal debt in the form of reverse mortgages or home equity loans—even at the advanced stage of the lifecycle—is yet another dimension to a healthy product allocation diet. Another product that certainly belongs in the optimal retirement portfolio is long-term care insurance, which is yet another part of a prudent risk management strategy.

Guarantees Make People Feel More Comfortable

Up until now I have discussed the rather theoretical models and processes for how people *should* be making decisions about their retirement products, pension annuities, and insurance riders. However, it's always interesting to see what investors and soon-to-be retirees are doing in practice, and so I conclude this chapter with a brief discussion of how people actually behave when it comes to their product and asset allocations.

Some convincing evidence emerges from a large database of variable annuity policyholders that the researchers at LIMRA International have been kind enough to allow us to probe and dissect (anonymously, of course). It turns out that when investors purchase guaranteed living benefits, the perception of these attained guarantees gives them the confidence to modify their investment behavior. Namely, they unequivocally accept greater equity market risk simply because they have this insurance; it remains to be seen how this will impact the market, availability, and pricing of these benefits in the long run.

Together with a graduate student of mine, we reviewed the overall asset allocation of variable annuity account holders and classified their investment asset allocations within these products into two